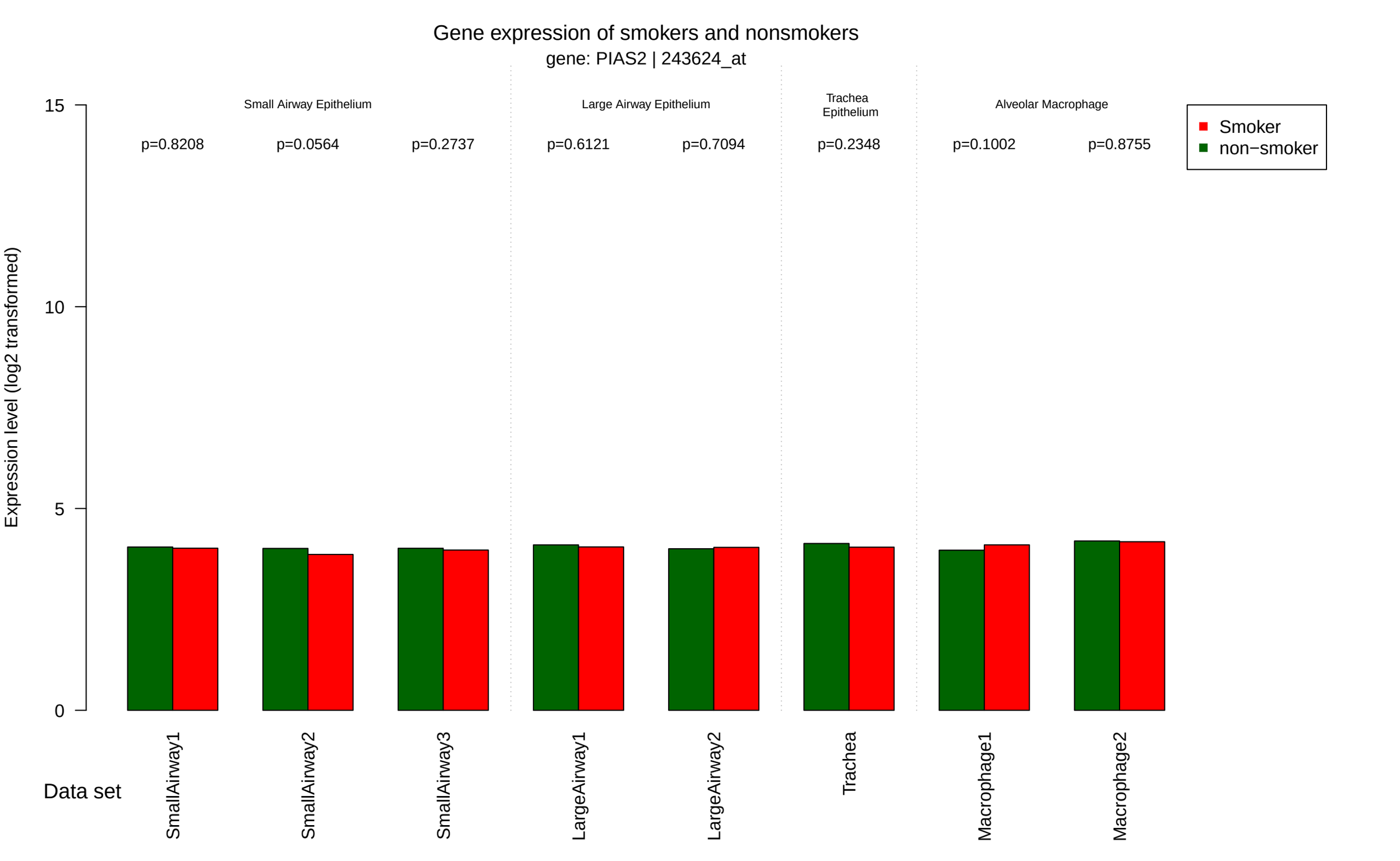
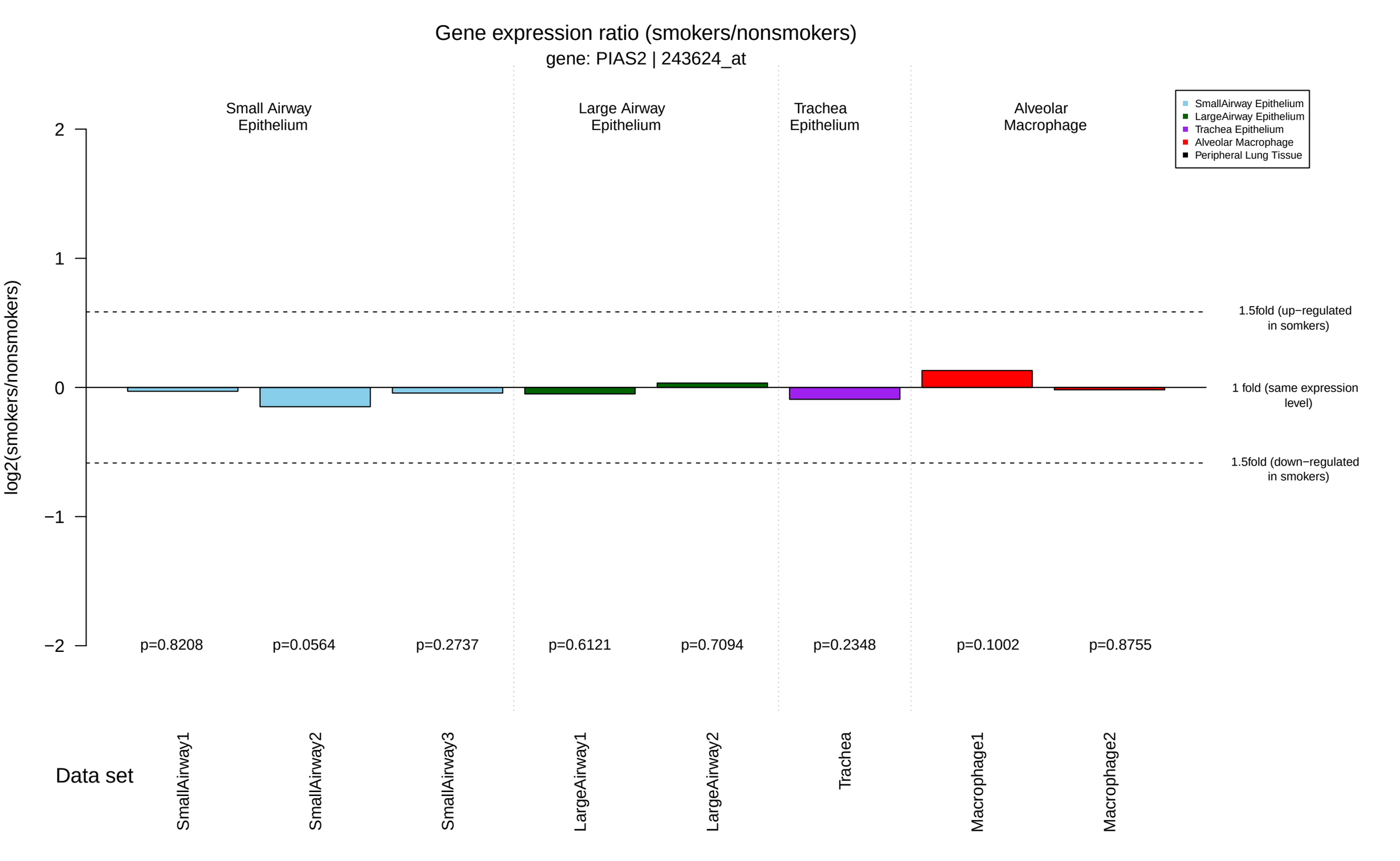
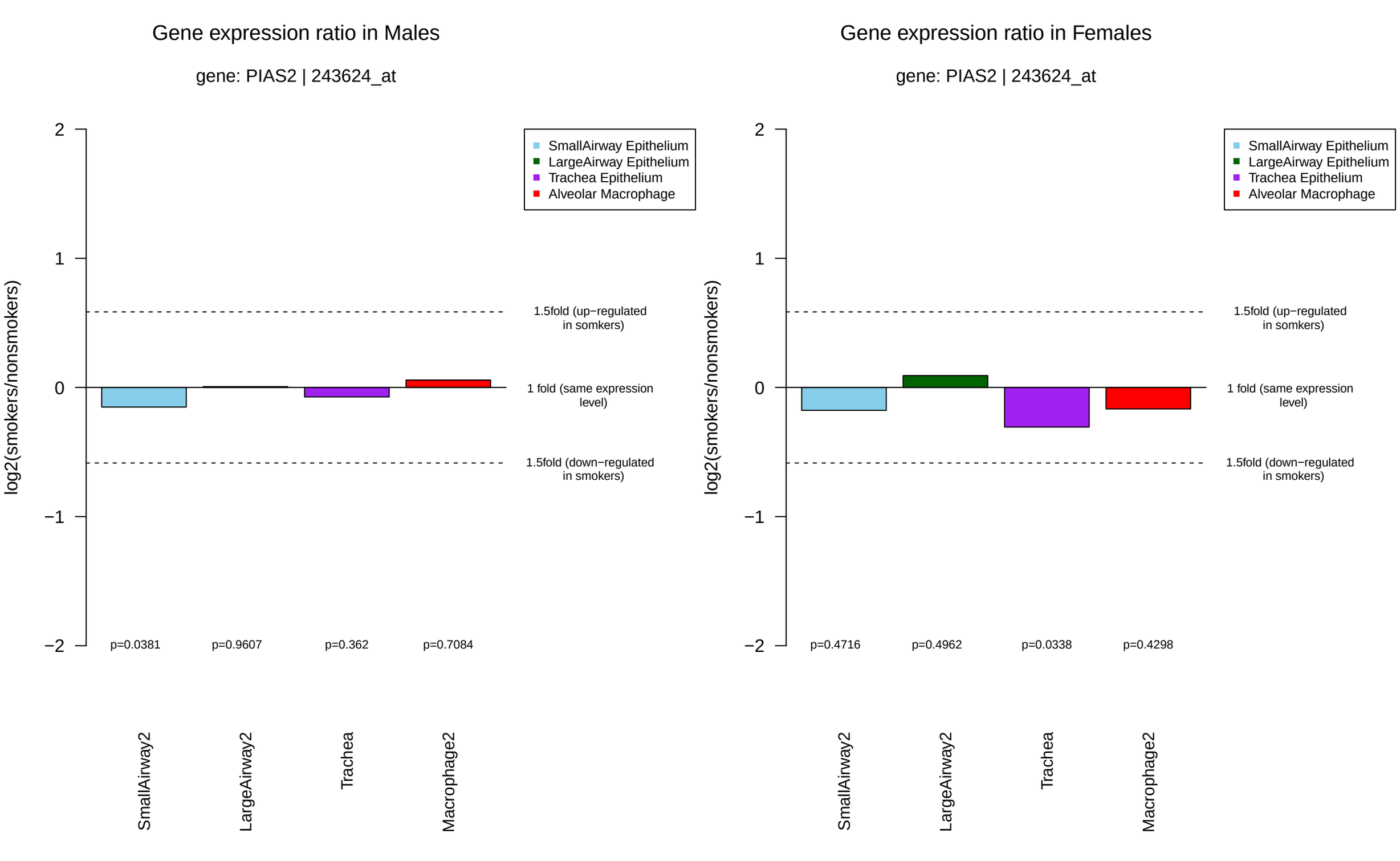
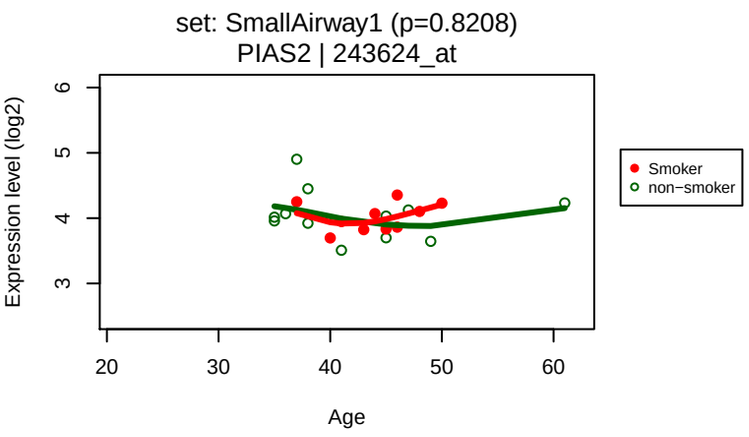
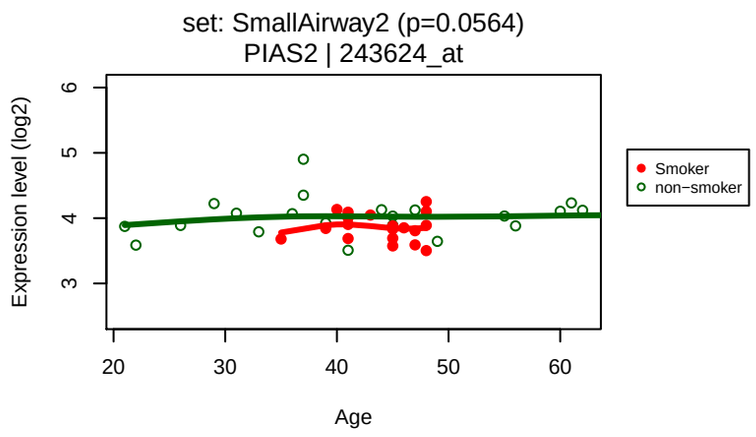
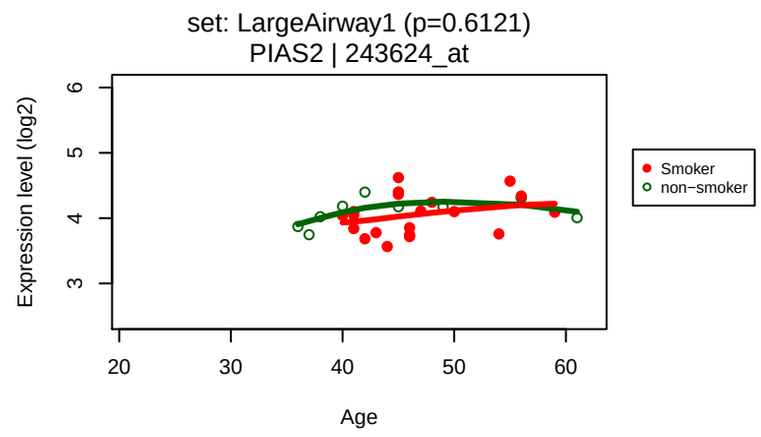

Gene expression of smokers and nonsmokers
gene: PIAS2 | 243624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8208
p=0.0564
p=0.2737
p=0.6121
p=0.7094
p=0.2348
p=0.1002
p=0.8755
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

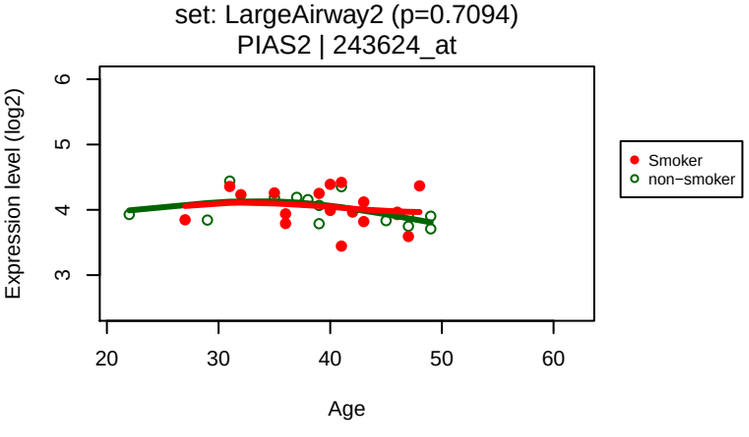

Gene expression ratio (smokers/nonsmokers)
gene: PIAS2 | 243624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8208
p=0.0564
p=0.2737
p=0.6121
p=0.7094
p=0.2348
p=0.1002
p=0.8755
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

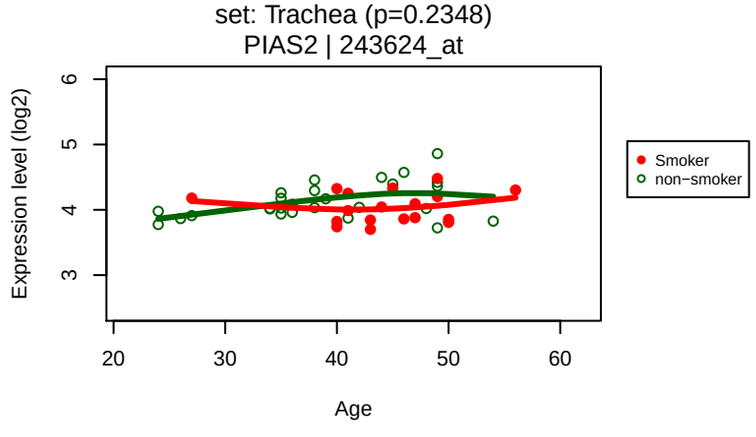

Gene expression ratio in Males
gene: PIAS2 | 243624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0381
p=0.9607
p=0.362
p=0.7084
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS2 | 243624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4716
p=0.4962
p=0.0338
p=0.4298
SmallAirway2
LargeAirway2
Trachea
Macrophage2

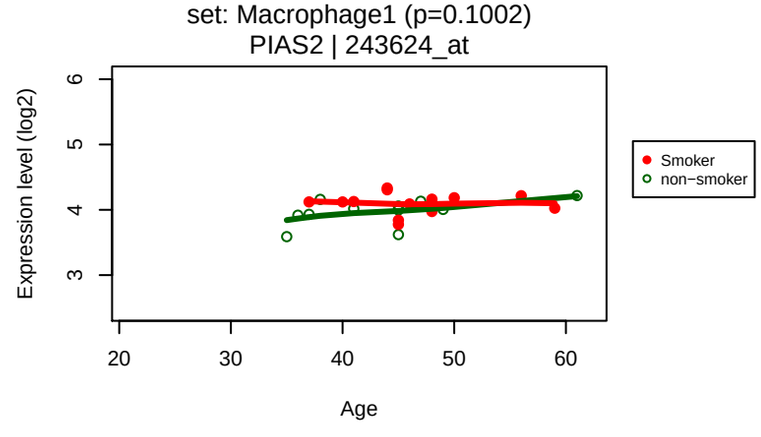

set: SmallAirway1 (p=0.8208)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker

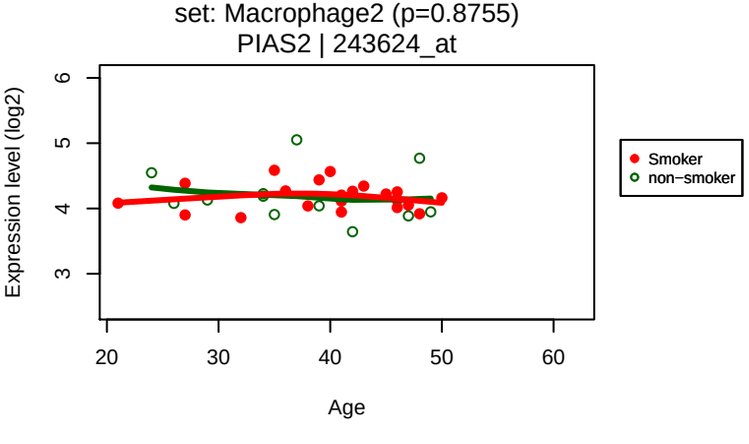

set: SmallAirway2 (p=0.0564)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker

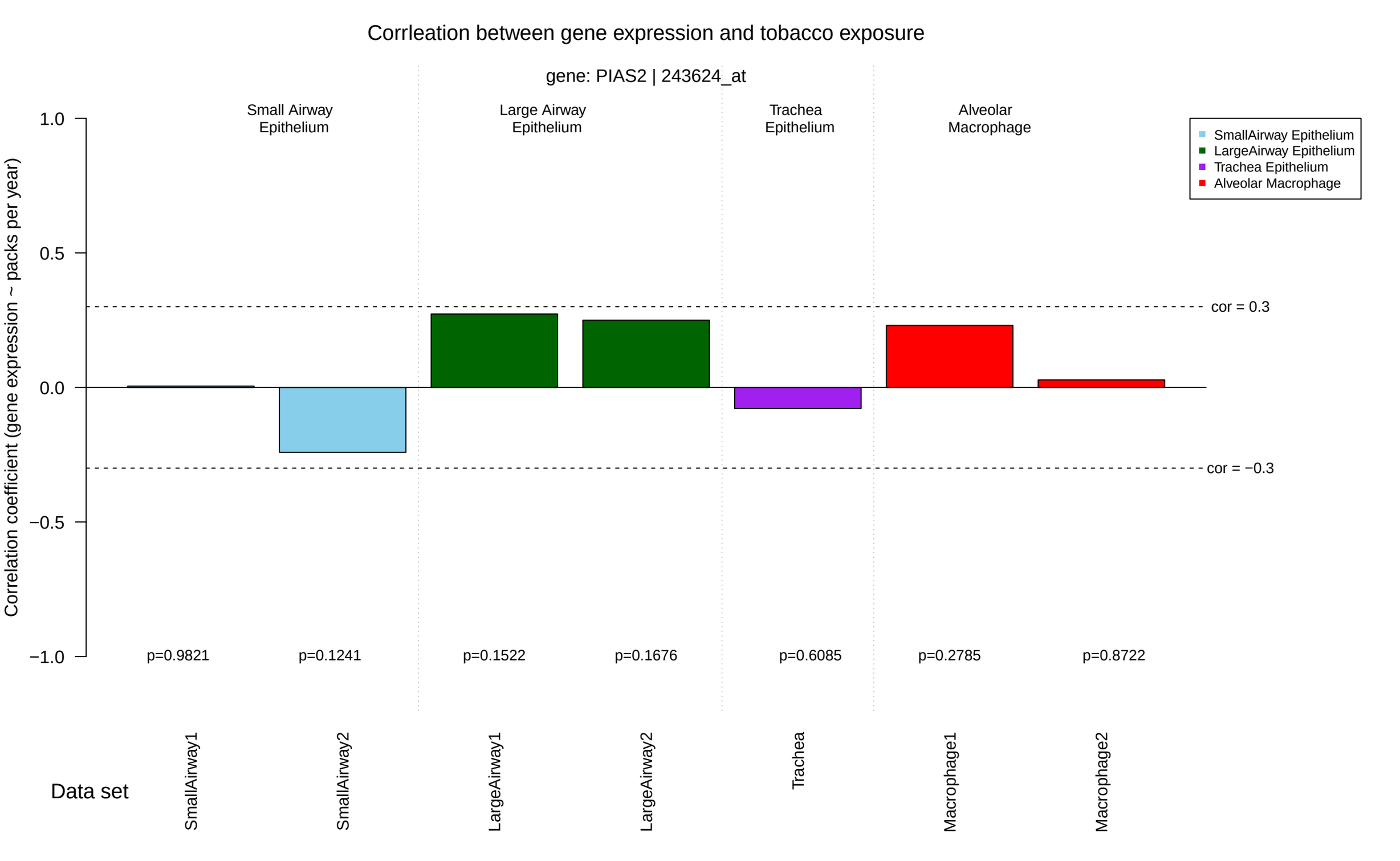

set: LargeAirway1 (p=0.6121)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7094)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2348)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1002)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8755)
PIAS2 | 243624_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS2 | 243624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9821
p=0.1241
p=0.1522
p=0.1676
p=0.6085
p=0.2785
p=0.8722
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2